

Engineer Role-Based Project Order

Evidence document for Professional Research Project 2

Document type	Evidence artifact
Portfolio collection	Professional Research Project 2
Decision supported	Choosing Fine-Tuning for Guarani Mbya

1. Purpose

This document supports my project choice for Professional Research Project 2. It orders the available project options based on how well they fit my GenAI Engineer role. This document is meant to be used as evidence for the related Decision Log, not as the Decision Log itself.

2. Role basis

The ordering is based on my GenAI Engineer profile. I prioritize projects that give evidence of technical investigation and model-level learning.

Higher ranking when the project includes	Lower ranking when the project mainly focuses on
Model behavior, adaptation, evaluation, experiments, benchmarks, measurable testing, and failure analysis.	UX polishing, creative media production, broad infrastructure, or API/tool integration without direct model improvement.

3. Ordering criteria

Criterion	Meaning in this project choice
Model-level relevance	The project involves model behavior, model improvement, evaluation, or adaptation.
Experimentation potential	The project allows tests, comparisons, measurable outputs, or technical evaluation.
New learning value	The project adds something new after my first Professional Research Project, StackEval.
Avoiding overlap	The project does not repeat the RAG/retrieval direction already present in my group project.
Realistic individual scope	The project is possible as an individual PRP2 prototype with limited time and student-level resources.

4. Project order list based on GenAI Engineer role

The table below shows the project order I used as evidence before writing the Decision Log.

Rank	Project	Engineer-role fit	Main technical focus	Selection note
1	Fine-Tuning for Guarani Mbya	Very strong	Fine-tuning; LoRA/QLoRA; limited data; memorization vs generalization	Selected: new model-level learning direction.
2	StackEval / When Leaderboards Lie	Very strong	LLM benchmarking; model comparison; human scoring; reliability	Not selected: already completed as PRP1.
3	KLM RAG Assistant	Strong	RAG; embeddings; chunking; retrieval quality; citations	Not selected: overlaps with group project workflow.
4	Ana Remembers	Strong / high-impact	Longitudinal memory; health tracking; pattern detection; escalation	Not selected: memory/health risk, not fine-tuning.
5	Betsy Procurement Agent	Medium-strong	Autonomous agents; planning; supplier memory; tool use	Not selected: agent automation, not model adaptation.
6	Parliament Speaks	Medium	MCP server; government APIs; tool design; caching	Not selected: mainly tool/API integration.
7	Reclamebureau Eva	Medium	Multi-agent orchestration; agent roles; marketing workflow	Not selected: more workflow/campaign oriented.
8	Zalando Virtual Models	Medium-low	Image-to-video; visual consistency; product fidelity	Not selected: visual/creative rather than fine-tuning.
9	Toys R Us Studios	Low for this role	Multi-modal video; voice; music; sound; editing pipeline	Not selected: least aligned with model-level engineering.

5. Why Fine-Tuning was selected over the other strong Engineer options

Why not StackEval again?

StackEval is highly aligned with the GenAI Engineer role because it focuses on LLM evaluation, benchmarking, human scoring, and model reliability. However, I already completed StackEval as my first Professional Research Project. Choosing it again would repeat my earlier learning instead of expanding my technical development for PRP2.

Why not KLM RAG Assistant?

KLM RAG is also a strong Engineer project because it focuses on retrieval quality, embeddings, chunking, citations, and hallucination prevention. However, this overlaps with my group project workflow, where I already work with retrieval, summarization, and quality evaluation. Choosing another RAG-focused project would repeat the same direction.

Why Fine-Tuning makes more sense for PRP2

Fine-Tuning gives me a different technical learning path. It moves from evaluating existing models or retrieving information toward adapting a model itself. This makes it a logical second project because it adds a new skill area: fine-tuning with limited data and testing whether the model generalizes or memorizes.

6. Research method note

Available Product Analysis is useful when investigating whether similar solutions already exist and whether it is worth recreating, embedding, or choosing one of them. In this project-choice context, I adapted that logic to compare available project options instead of commercial products. This helped me identify overlap with work I already completed or was already doing, especially StackEval and RAG.

Conclusion: Fine-Tuning for Guarani Mbya is the best PRP2 choice because it is strongly aligned with the GenAI Engineer role, adds new model-level learning after StackEval, avoids repeating the group project RAG workflow, and provides clear technical evidence through dataset preparation, fine-tuning, demo output, and evaluation.